

CYBR 437 SECURE CODING

Syllabus - Fall 2026

General Information:

- Lecture Time and Location: Monday, Wednesday, Friday, 9:00 – 9:50 am in Catalyst Building, Room 220
- Lab Time and Location: Thursday, 2:00 – 3:50 pm in Catalyst Building, Room 193
- Course Canvas site: <https://canvas.ewu.edu/courses/1925348>

Instructor:

Abinash Borah

Email: aborah@ewu.edu

Office: Catalyst Building, Room 129

Office Hours: Monday 10 – 11 am; Wednesday and Friday 10 – 11:30 am

(and other times through appointments made by email or Canvas message)

Teaching Assistant:

Paul Aguilar

Email: paguilar4@ewu.edu *(If you need to email the TA, please CC the instructor)*

Office: Catalyst Building, Room 139

Office Hours: Monday – Thursday, 12 – 1 pm

Course Description: This course introduces a variety of topics of concern to programmers when writing code, examines the characteristics of secure programs and the ability to implement programs that are free from vulnerabilities, including evaluating software, understanding mechanisms for testing software for vulnerabilities, and understanding specific vulnerabilities such as buffer overflow.

Learning Outcomes: After completing this course, students will be able to:

1. Describe the characteristics of secure programming
2. Recognize the importance of secure software and programming practices, and development processes and methodologies that lead to secure software
3. Identify the vulnerabilities inherent in different programming languages
4. Examine vulnerabilities introduced by specific programming language libraries and how to mitigate those vulnerabilities
5. Apply software security analysis tools and techniques

Companion Course:

CYBR 437L Secure Coding Lab (to be assessed and graded together, and the same grade will be awarded for both courses)

Code of Ethics and Professional Conduct:

Students are expected to uphold a code of ethics and professional behavior that promotes the highest standards of integrity, honesty, trustworthiness, and professionalism when exploring cybersecurity-related topics and exercises. Students are required to sign the NCAE-C Student Code of Ethics and Professional Conduct during the first week of class to proceed in the course. A copy of the same can be obtained by emailing cybersecurity@ewu.edu. Printed copies will be provided in class for students to complete and return to the instructor by Wednesday, September 30th.

Textbooks: No textbooks are required for this course. The instructor will provide any required material in addition to the lecture slides as the course progresses.

Course Schedule:

The following schedule is subject to change (*though I will try my best to avoid changes*). Any changes made will be notified via Canvas and in-class announcements, and an updated syllabus will be posted in Canvas.

Week	Topics	Assignment Release	Assignment Due
1 Sep 23 – 25	Introduction	Lab-1: Sep 24	-
2 Sep 28 – Oct 2	C Review	Lab-2: Oct 1	Lab-1: Sep 30 Quiz-1: Oct 1 (in lab)
3 Oct 5 – 9	Vulnerability Assessment	Lab-3: Oct 8	Lab-2: Oct 7
4 Oct 12 – 16	Regular Expressions Secure Coding in C	Lab-4: Oct 15	Lab-3: Oct 14 Quiz-2: Oct 15 (in lab)
5 Oct 19 – 23	Buffer Overflow	Lab-5: Oct 22	Lab-4: Oct 21
6 Oct 26 – 30	Buffer Overflow (contd.)	Lab-6: Oct 29	Lab-5: Oct 28 Midterm: Oct 30 (in class)
7 Nov 2 – 6	Buffer Overflow (contd.)	Lab-7: Nov 5	Lab-6: Nov 4 Quiz-3: Nov 5 (in lab)
8 Nov 9 – 13	Secure Coding in Java and Python	Lab-8: Nov 12	Lab-7: Nov 12
9 Nov 16 – 20	Rust for Security	Lab-9: Nov 19	Lab-8: Nov 18 Quiz-4: Nov 19 (in lab)
10 Nov 23 – 24	Security of AI Applications	-	Lab-9: Nov 24
11 Nov 30 – Dec 4	Security of AI Applications (contd.)	-	Quiz-5: Dec 3 (in lab)
12 Dec 7 – 10	Final Exam	-	Final Exam: Dec 8, 8-10 am (in class)

Assessment of Learning Outcomes:

Students' achievement of the learning outcomes of this course will be measured using the following types of assignments. Detailed assignment information and rubrics will be posted in

Canvas in due course. The assignments will be evaluated for both completion and correctness. If an assignment is submitted multiple times, the latest submission will be graded.

- **Quizzes:** There will be in-lab paper-based quizzes to assess students' understanding of the course material and their ability to apply it in solving problems. Quiz questions will be based on the topics being covered at the respective time. The lowest quiz grade will be excluded when calculating students' total grade.
- **Labs:** Labs will be assigned to assess students' grasp of the course material and their ability to apply it in hands-on exercises and solving conceptual problems.
- **Exams:** There will be one midterm and one final exam. Both exams will be in-class and paper-based.

Grading Scale:

The weight distribution for the different types of assignments for calculating the total grade will be as follows:

Assignment Type	Percentage of Total Grade
Quizzes	25%
Labs	45%
Midterm Exam	15%
Final Exam	15%

The total grade will be rounded up to the next whole percentage, and a final letter grade will be reported on the following scale:

Total Grade	Letter Grade
$\geq 92\%$	A
88% - 91%	A-
84% - 87%	B+
81% - 83%	B
78% - 80%	B-
75% - 77%	C+
72% - 74%	C
69% - 71%	C-
66% - 68%	D+
63% - 65%	D
60% - 62%	D-
$< 60\%$	F

The instructor reserves the right to change the above percentages to adjust grades beyond this rubric.

Course Policies:

- **Attendance:** Professional behavior is expected throughout the course. This includes, but is not limited to, attendance, punctuality, constructive class participation, and effectively communicating with me.

Class participation is important, including class attendance. It is expected that you will complete any assigned work in time and be prepared for class. Absences and/or not participating in class will negatively impact your learning, and you will be accountable for the material covered in class. If you need to miss class for unavoidable reasons, please ensure that you contact your classmates or me to catch up on the material. Disruptive behavior that unreasonably interferes with my ability to teach or other students' ability to learn, such as engaging in private conversations or using electronic devices for non-class activities, is highly discouraged.

- **Late Work:** It is important that you meet course deadlines and complete assignments in time to participate in class discussions and activities. Late work will receive a 10% late penalty per day unless an extension is granted by me for valid reasons. However, no late work will be accepted beyond three days past the due date (with a 30% late penalty). If you have a professional opportunity (e.g., a conference), medical situation (including caring for dependents, etc.), or any other valid reason that may cause an assignment to be submitted late, please contact me in advance to request an extension.

If you face technical issues while submitting an assignment in Canvas, email me with a description of the issue and attach the assignment before the deadline to avoid it being considered late. If needed, contact the IT help desk (Email: helpdesk@ewu.edu) to resolve the technical issue and upload the assignment to Canvas as soon as possible.

- **Makeup Assignments:** There will be no make-up quizzes and exams without my prior approval. Such prior approval will require valid reasons from the students, preferably supported with some documentation.
- **AI Policy:** Unless otherwise specified in an assignment, the use of AI for completing any form of assignment in this course is prohibited.
- **Communication:** I will communicate with you outside of class using Canvas announcements and Canvas messages. Please regularly check your email (and Canvas, if notifications on announcements and messages in Canvas are not forwarded to your email). I will also make regular in-class announcements.
Please send emails or Canvas messages to reach out to me outside of class. If you send me an email, please include the course number. I will usually reply within 12 hours to emails and messages received Mondays through Fridays between 8 am and 5 pm. Emails and messages received after 5 pm or at weekends may not be read until the next working day.

University Policies:

- **Academic Integrity:** Academic Integrity is the cornerstone of the university. Any student who attempts to gain an unfair advantage over other students by violating the Academic Integrity policy may be reported to the university and may receive a sanction up to and including an XF grade for the course, suspension, or expulsion from the university. This

policy is on the EWU website at <https://inside.ewu.edu/policies/knowledge-base/wac-172-90-student-academic-integrity-3/>

Examples of academic dishonesty include misrepresenting others' work as your own, failure to provide appropriate citations in written text, stealing or destroying material intended for the use of other students, using notes or cell phones during exams, taking photographs of exams, assisting others during exams, copying someone else's work, or letting your own work be copied.

- **Disability Support and ADA:** Eastern Washington University is committed to providing support for students with disabilities. If you are a student with physical, learning, emotional, or psychological disabilities needing accommodation, you are encouraged to contact Student Accommodations and Support Services (<https://inside.ewu.edu/sass/>). If you have approved accommodation selected for this course, I will reach out to you within the second week of the quarter to initiate a discussion on implementing your accommodation in this course.
- **Title IX & Sexual Misconduct:** As an instructor, one of my responsibilities is to help create a safe learning environment. I also have a mandatory reporting responsibility related to my role as a faculty member. It is my goal that you feel able to share information related to your life experiences in this class. I will seek to keep information you share private to the greatest extent possible. However, I am required to share information regarding sexual misconduct with the University Title IX Coordinator (Email: titleix@ewu.edu). Students may speak to someone confidentially and find support via resources found here: <https://inside.ewu.edu/titleix/services/confidential-services/>
- **Religious Accommodations:** If you would like to request an accommodation for reasons of faith or conscience, please refer to EWU's policy on Holidays and Religious Accommodations (<https://inside.ewu.edu/policies/knowledge-base/ewu-403-01-holidays/>) and submit a request using the form linked here within the first two weeks of the quarter: <https://inside.ewu.edu/studentaffairs/holidays-and-religious-accommodations/>
- **Other University Policies:** Please refer to the Online Student Handbook at <https://inside.ewu.edu/studentaffairs/student-handbook/>

Student Support Services:

- **Student Affairs** for resources on course support, student advocacy with course or self-care challenges, childcare, counseling and wellness, and financial support, including the free food pantry and student emergency fund: <https://inside.ewu.edu/studentaffairs/resources-for-students/>
- **IT Help Desk** for support with tech-related issues related to Canvas or course activities: <https://support.ewu.edu/support/home>
- **EWU Library** for support with research, class materials, journal articles, etc.: <https://www.ewu.edu/library>

- **Writers' Center** for guidance on writing skills, grammar, citations, etc., including grammar and writing guides, online workshops, and appointments to review your writing: <https://inside.ewu.edu/writerscenter/>
- **Veterans Resource Center** for support navigating the transition from military to academic life, VA education benefits, and community: <https://inside.ewu.edu/veterans/>
- **PLUS Group** for study groups, tutoring, and academic coaching: <https://inside.ewu.edu/plus/>